

RAGHAV BAJAJ

Los Angeles, CA | +12138204318 | raghav.bjj77@gmail.com | [LinkedIn](#) | [Github](#)

Summary

Entry-Level Python Developer with practical experience in building scalable backend services and automating CI/CD pipelines. Proficient in Python and familiar with web frameworks, with hands-on experience in integrating user-facing elements with server-side logic. Proven ability to enhance application performance and maintain code quality through rigorous testing and collaboration within Agile teams.

Education

University of California, Riverside

Sep 2023 - Jun 2025

Master of Science, Computer Science & Engineering

- **GPA:** 3.7/4.0
- **Coursework:** Design and Analysis of Algorithms, Data Mining, Machine Learning, Real Time Embedded Systems

Gujarat Technology University, Gujarat

Jun 2019 - May 2023

Bachelor in Engineering, Computer Science & Engineering

- **GPA:** 3.8/4.0
- **Achievements:** Captain of Badminton Team

Experience

Centre for Human Services

Mar 2025 - Present

Software Engineer

Modesto, CA

- Engineered a secure Spring Boot-based backend system to replace Excel-based tracking for the Youth Navigation Center (YNC), enabling centralized client data management and streamlined agency-wide operations.
- Developed infrastructure around the core Party (Client) entity to support lifecycle tracking across departments, improving visibility and reducing duplication.
- Collaborated with data and IT teams to implement scalable enhancements using Google Cloud Platform and PostgreSQL, bolstering data integrity and facilitating smooth onboarding for non-technical stakeholders.

ASSYST Inc

Jun 2024 - Sep 2024

Software Engineer Intern

Sterling, VA

- Executed full stack application deployment in Kubernetes, emphasizing CI/CD pipeline automation to improve deployment efficiency and minimize manual errors.
- Constructed a CI pipeline with Tekton to automate repository cloning, Docker image creation, and deployment via Argo CD.
- Integrated SonarQube and Grype to uphold code quality and perform vulnerability scanning for secure software deployments.
- Developed a resume matcher web app by leveraging React with Typescript for UI development and Flask for backend services, thereby enhancing resume processing and matching efficiency with scalable Python code.
- Engineered API calls to compare job descriptions of uploaded resumes, refining the accuracy and relevance of job matching.

Unity Technology

Jan 2023 - Aug 2023

Software Engineer Intern

Surat, India

- Collaborated with a cross-functional team to design and maintain Flutter-based mobile applications for Android and iOS platforms, contributing to a 20% increase in user engagement.
- Implemented a real-time chat interface, enabling dynamic user interactions with the chatbot and significantly enhancing user satisfaction.
- Integrated an NLP engine to process user input, accurately interpret intents, and deliver pertinent responses, thereby boosting the chatbot's performance and reliability.

Projects

NYC For-Hire Vehicle Big Data Analysis | [Link](#)

Jan 2024 - Mar 2024

- Analyzed NYC For-Hire Vehicle data using Apache Spark and Python to extract key mobility trends and zone patterns for data-driven transportation planning.
- Visualized peak vs off-peak patterns and key metrics in Jupyter Notebooks.
- Identified trends in urban mobility, trip durations, and fare structures, supporting targeted transportation planning.

Forecasting Credit Ratings in the Banking Sector | [Link](#)

Sep 2023 - Dec 2023

- Developed a model to forecast credit ratings, mitigating loan default risks through financial indicator analysis.
- Created Tableau dashboards to visualize trends, assisting financial institutions in data-driven decision-making.

Black Friday Sales Trends Analysis | [Link](#)

Sep 2024 - Nov 2024

- Built an XGBoost model using Python and Pandas to predict high spenders, achieving a recall of 0.98 for customer segmentation.
- Conducted data preprocessing and feature engineering (e.g., Age $\times$ City) to uncover spending patterns and optimize insights.
- Visualized key trends using Matplotlib, helping translate model findings into business insights.

AI-Powered Pacman Navigation Agent Development

Mar 2024 - May 2024

- Engineered search algorithms(BFS,DFS,UCS,A*), reducing search nodes from 2455 to 910 for efficiency.

Skills

- **Programming Languages:** JAVA, C, C++, Python, SQL
- **DevOps/CI CD Tools:** Kubernetes, Jenkins, Tekton, Docker, Argo CD, SonarQube, Grype, GitLab
- **Frameworks:** JavaScript, React.js, Angular.js, Django, Node.js, HTML, CSS, Hadoop
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Cloud:** Google Cloud Platform, Amazon Web Services
- **UI UX Tools:** Figma, Adobe XD